

1. Consider the function $f(x, y) := |x| + 2|y|$.
 - (a) Find all the numbers k such that the level curve $L_k := \{(x, y) \in \mathbb{R}^2 : f(x, y) = k\}$ is not an empty set.
 - (b) Sketch the curves $L_k \neq \emptyset$.

2. Evaluate the limit or show that it does not exist:
 - (i) $\lim_{(x,y) \rightarrow (0,0)} \frac{x \sin y}{y \sin x}$;
 - (ii) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$;
 - (iii) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y}{y^2 + x}$.

3. Consider the function $f(x, y) := \begin{cases} \frac{\sin^3 x}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$
 - (a) Study the continuity of f at every $(x, y) \in \mathbb{R}^2$.
 - (b) Find, if they exist, $f_x(0, 0)$ and $f_y(0, 0)$.

4. Sketch the level curves and the graphs of the following functions:
 - (a) $f(x, y) = \frac{1}{x^2 + y^2}$;
 - (b) $f(x, y) = x^2 + 3y^2$;
 - (c) $f(x, y) = e^{-x^2 - y^2}$.

5. Given the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = \frac{x + y}{2x^2 + 2y^2 + 4}$ for every $(x, y) \in \mathbb{R}^2$.
 - (i) Describe the level curves of the function f . (Hint: A similar question is found in the notes).
 - (ii) Describe the level curve passing through the point $(0, 1)$.
 - (iii) Find both partial derivatives $f_x(1, -1)$ and $f_y(1, -1)$ of the function f at the point $(1, -1)$.
 - (iv) Write down the cartesian equation of the tangent plane to the graph of f at the point $(1, -1)$.